Lab Exercise 17- Creating Secrets in Kubernetes

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssw0rd'
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

The output will be base64 encoded. To decode, you can use:

```
echo -n “P@ssw0rd” | base64
```

PowerShell Command: -

```
[Convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes("P@ssw0rd"))
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl create secret generic my-secret --from-literal=username=admin --from
-literal=password='P@ssw0rd'
secret/my-secret created
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-15T16:40:24Z"
  name: my-secret
  namespace: default
  resourceVersion: "5365"
  uid: f0034adf-154d-4f90-8286-1f0b786ea01a
type: Opaque
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> wsl
docker-desktop:/tmp/docker-desktop-root/run/desktop/mnt/host/c/Users/Devanshi/Desktop/COS_Lab/lab17# echo -n “P@ssw0rd”
| base64
4oCcUEBzc3cwcmTigJ0=
docker-desktop:/tmp/docker-desktop-root/run/desktop/mnt/host/c/Users/Devanshi/Desktop/COS_Lab/lab17# exit
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> [Convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes("P@ssw0rd"))
UEBzc3cwcmQ=
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

```
echo -n 'my-username' > ./username.txt
echo -n 'my-password' > ./password.txt
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> echo -n 'my-username' > ./username.txt
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> echo -n 'my-password' > ./password.txt
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl create secret generic my-file-secret --from-file=username=./username
.txt --from-file=password=./password.txt
secret/my-file-secret created
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
  password: //5tAHkALQBwAGEAcwBzAHcAbwByAGQADQAKAA==
  username: //5tAHkALQB1AHMAZQByAG4AYQBtAGUADQAKAA==
kind: Secret
metadata:
  creationTimestamp: "2026-03-15T16:42:12Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "5509"
  uid: 48043d64-911a-4ac1-be46-f01210bd043d
type: Opaque
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17>
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

```
kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
```

Verify the secret:

```
kubectl get secrets my-registry-secret -o yaml
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl create secret docker-registry my-registry-secret --docker-username=<
your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
secret/my-registry-secret created
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl get secrets my-registry-secret -o yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJcdTAwM2N5b3VyLWRvY2tlci1zZXJ2ZXJcdTAwM2UiOnsidXNlcm5hbWUiOiJcdTAwM2N5b3VyLXVzZXJuYW1
lXHUwMDNlIiwicGFzc3dvcmQiOiJcdTAwM2N5b3VyLXBhc3N3b3JkXHUwMDNlIiwiZW1haWwiOiJcdTAwM2N5b3VyLWVtYWlsXHUwMDNlIiwiYXV0aCI6IlB
IbHZkWEl0ZFhObGNtNWhiV1UrT2p4NWIzVnlMWEJoYzNOM2IzSmtQZz09In19fQ==
kind: Secret
metadata:
  creationTimestamp: "2026-03-15T16:43:31Z"
  name: my-registry-secret
  namespace: default
  resourceVersion: "5619"
  uid: 994ae1d7-7186-410d-a742-96c612ca28a2
type: kubernetes.io/dockerconfigjson
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17>
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
```

Apply the YAML file:

```
kubectl apply -f my-secret.yaml
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

```
my-secret.yaml
my-secret.yaml
1  apiVersion: v1
2  kind: Secret
3  metadata:
4    name: my-manifest-secret
5  type: Opaque
6  data:
7    username: YWRtaW4=    # base64 encoded value of 'admin'
8    password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
9
10
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl apply -f my-secret.yaml
secret/my-manifest-secret created
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-15T16:40:24Z"
  name: my-secret
  namespace: default
  resourceVersion: "5365"
  uid: f0034adf-154d-4f90-8286-1f0b786ea01a
type: Opaque
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

pod-with-secret.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl apply -f pod-with-secret.yaml
pod/pod-with-secret created
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> wsl
docker-desktop:/tmp/docker-desktop-root/run/desktop/mnt/host/c/Users/Devanshi/Desktop/COS_Lab/lab17# kubectl exec pod-wi
th-secret -- printenv | grep -E 'USERNAME|PASSWORD'
-sh: kubectl: not found
docker-desktop:/tmp/docker-desktop-root/run/desktop/mnt/host/c/Users/Devanshi/Desktop/COS_Lab/lab17# minikube start
-sh: minikube: not found
docker-desktop:/tmp/docker-desktop-root/run/desktop/mnt/host/c/Users/Devanshi/Desktop/COS_Lab/lab17# exit
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
grep : The term 'grep' is not recognized as the name of a cmdlet, function, script file, or operable program. Check
the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:44
+ kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD ...
+                                            ~~~~
    + CategoryInfo          : ObjectNotFound: (grep:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl exec pod-with-secret -- printenv | Select-String "USERNAME|PASSWORD"


USERNAME=admin
PASSWORD=P@ssw0rd
```

Step 7: Clean Up

Delete the secrets:

```
kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
```

Delete the pod:

```
kubectl delete pod pod-with-secret
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl delete secret my-secret
secret "my-secret" deleted
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl delete secret my-file-secret
secret "my-file-secret" deleted
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl delete secret my-registry-secret
secret "my-registry-secret" deleted
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl delete secret my-manifest-secret
secret "my-manifest-secret" deleted
PS C:\Users\Devanshi\Desktop\COS_Lab\lab17> kubectl delete pod pod-with-secret
pod "pod-with-secret" deleted
```